

Salt Basin · Tech Stack Architecture

SALT BASIN NET WORKS · BOTTOM LINES WITH A RISING TIDE

SYSTEM OVERVIEW

Tech Stack Architecture

saltbasin.net is delivered by five composable layers — presentation, application, data, external services, and infrastructure. The total stack costs **\$0/month** at current volume (free tiers across the board). The diagram below shows the layered architecture; the inventory tables document the specific tools at each layer.

LAYERED ARCHITECTURE

WHAT THE VISITOR SEES

Presentation Layer

React 19 Vite React Router CSS variables Cormorant Garamond + Inter SVG (custom industry wheel)
CSS @media print CSS scroll-snap

Single React bundle served by Netlify CDN. Admin, member admin, public profile, lead view, and print-friendly outputs are all the same SPA with different routes.

SERVER-SIDE BUSINESS LOGIC

Application / Domain Layer

Express 4 Node 22 bcryptjs (password hashing) cookie-parser multer (uploads) cors
middleware: requireAdmin / requireUser / requireLeadAccess libsodium-wrappers (GitHub secret encryption)

Stateless Express app on Render. All routes under /api/*. Three auth scopes (admin / member / lead) layered as middleware.

PERSISTENCE

Data Layer

Supabase Postgres 15 postgres (npm pkg, async pool) Supabase Storage (uploaded images)
JSON-in-TEXT for flexible state (config, sites) idempotent ALTER TABLE migrations on boot

13 tables: users, sessions, site_state, config_state, leads, lead_emails, lead_activity, lead_sessions, member_profiles, member_sites, member_configs, capability_groups, backlog_items, tier_workarounds. Migrations are idempotent and applied on every boot.



THIRD-PARTY INTEGRATIONS

External Services

Brevo (transactional email) Zoho Mail (inbound) Wix DNS (DNS authority) Anthropic API (BYO key, member-side)
GitHub (source of truth + Actions CI) Render API (deploy automation) Netlify API (deploy automation)
UptimeRobot (cold-start pinger)

Pure HTTPS, no SDKs except Anthropic. Every external call has a fallback (console stub for email, in-memory for cache).

DEPLOYMENT + OBSERVABILITY

Infrastructure

Render (Express backend host) Netlify (static + CDN + proxy) Supabase (managed Postgres)
GitHub Actions (CI + deploy verify + secret rotation) Wix (registrar + DNS) UptimeRobot (free uptime pings)

Six free-tier products composed to deliver a production website with zero monthly fixed cost. Netlify proxies /api/* to Render via netlify.toml.

TECH BY CAPABILITY

What each capability is built from

Platform Foundation

React 19 Vite React Router Express Postgres (Supabase) bcryptjs cookie-parser CSS variables

Multi-tenant CMS

React 19 Express Postgres (JSONB-style TEXT) AdminShell shared component requireUser middleware

Lead Capture & Identity

React 19 Express Postgres bcryptjs (lead passwords) crypto (publicid generation) lead_sessions cookies

Email Infrastructure

Brevo API (v3/smtp/email) Zoho Mail (Forever Free) Wix DNS SPF + CNAME-based DKIM + DMARC fetch (no SDK)
lead_emails audit table

Salt Basin Net Works

React 19 Express Postgres CSS scroll-snap (horizontal banner) one-shot script (inject-networks-banner.mjs)



Public Site Content

React 19

SVG (industry wheel)

CSS Grid + Flexbox

Cormorant Garamond + Inter web fonts

Output Pages (Resume / Proposal / Case Study / One-Pager)

React 19

CSS @media print

GatedPreview component (auth gate)

OutputFrame wrapper

Admin Experience

React 19

multer (image uploads)

Supabase Storage

CSS variables (light/dark adaptive)

mobile breakpoint <768px

Deployment Infrastructure

Render (free tier)

Netlify (free tier)

Supabase (free tier: Postgres + Storage)

GitHub Actions

libsodium-wrappers (encrypted GH secrets)

Render API

Netlify API

Security & Data

bcryptjs

node:crypto (randomBytes)

cookie-based sessions

rotate-admin-password.mjs script

gitignored secret files

Observability & Quality

GitHub Actions

UptimeRobot (13-min pings)

Render API polling

auto-issue creation on failure

netlify.toml build.ignore script

Requirements & Test Management

React 19

Express

Postgres (capability_groups + backlog_items + tier_workarounds)

side-drawer pattern

auto-seed on first load

Generated live from the admin Backlog · 69 delivered requirements

